Lecture 16-17

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1 Relations

1.1 Definition

Let A and B be sets. A relation from A to B is a subset $R \subset A \times B$.

Notation:

- $(a, b) \in R \iff aRb$
- $(a, b) \notin R \iff a \not R b$

If $A = B$, then R is called a relation on A.

1.2 Examples

1.2.1 Example 1: Relation between distinct sets

Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$. Let $R = \{(1, a), (1, c), (2, b), (2, c), (3, c)\}$. This can be represented by a mapping diagram where arrows point from elements in A to elements in B.

1.2.2 Example 2: Inequality relation on a set

Let $R \subset A \times A$ where $A = \{1, 2, 3, 4\}$. Let $aRb \iff a < b$. Then $R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$. Note: In the scanned text, there seems to be a mistake listing $(4, 2), (4, 3)$ etc., which would correspond to $>$ rather than $<$. Given the context of standard relations, $a < b$ usually implies the pairs listed above.

1.2.3 Example 3: Identity/Diagonal relation

Let $A = \{1, 2, 3, 4\}$. Let $aRb \iff a = b$. Then $R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$.

2 Properties of Relations

Let R be a relation on a set S.

2.1 1. Reflexive

R is reflexive if for all $a \in S$, aRa.

- Example of reflexive relation: $R_1 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$ on $\{1,2,3,4\}$. Wait, looking closely at R_1, $(3,3)$ is missing, so R_1 is actually **not** reflexive on $\{1,2,3,4\}$.
- R_5 (from the table) contains $(1,1),(2,2),(3,3),(4,4)$, so it is reflexive.

2.2 2. Symmetric

R is symmetric if for all $a, b \in S$, $aRb \implies bRa$.

- Examples from text: R_2 and R_3 are symmetric.
- Note: $R_2 = \{(1,1),(1,2),(2,1)\}$. Since $(1,2) \in R_2 \implies (2,1) \in R_2$, it is symmetric.

2.3 3. Antisymmetric

R is antisymmetric if for all $a, b \in S$, $(aRb \wedge bRa) \implies a = b$.

- Examples from text: R_4, R_5, R_6.
- R_4 represents a strictly less than or similar partial order relation where no two distinct elements relate to each other both ways.

2.4 4. Transitive

R is transitive if for all $a, b, c \in S$, $(aRb \wedge bRc) \implies aRc$.

- Examples from text: R_4, R_5, R_6.

2.5 Counting Relations

If $|S| = n$, then $|S \times S| = n^2$. The number of possible relations on S is the number of subsets of $S \times S$, which is $|\mathcal{P}(S \times S)| = 2^{n^2}$.

3 Equivalence Relations

3.1 Definition

A relation R on a set S is called an **equivalence relation** if R is:

1. Reflexive
2. Symmetric
3. Transitive

Notation: $a \sim b$ is often used for equivalence relations.

3.2 Examples

1. $S = \mathbb{Z}$, $aRb \iff |a| = |b|$. (Reflexive, Symmetric, Transitive -> YES)
2. $S = \mathbb{R}$, $aRb \iff |a - b| \leq 2$. (Reflexive? Yes. Symmetric? Yes. Transitive? No, e.g., $0R1.5$ and $1.5R3$ but not $0R3$. -> NOT an equivalence relation).
3. $S =$ PEOPLE, $aRb \iff a$ is a sibling of b. (Transitivity fails if you don't count oneself as a sibling, or if you consider half-siblings in a specific way. Usually NOT transitive in strict definitions without care).
4. $S =$ CITIES, $aRb \iff a$ is in the same country as b. (YES, this partitions cities by country).
5. Modular Arithmetic: $S = \mathbb{Z}$, fix $m \in \mathbb{N}, m \geq 2$. Define $aRb \iff a \equiv b \pmod{m}$. (YES).

4 Equivalence Classes

4.1 Definition

Let R be an equivalence relation on S and $a \in S$. The set

$$[a] = \{s \in S : (a, s) \in R\}$$

is called the **equivalence class** of a.

4.2 Theorem (Properties of Equivalence Classes)

Let R be an equivalence relation on S, and $a, b \in S$. Then the following are equivalent (TFAE):

1. aRb
2. $[a] = [b]$
3. $[a] \cap [b] \neq \emptyset$

4.2.1 Proof

- $(i) \implies (ii)$: Assume aRb. We want to show $[a] \subset [b]$ and $[b] \subset [a]$. Let $x \in [a]$, so aRx. Since R is symmetric, bRa. Since R is transitive and $bRa \wedge aRx$, we have bRx, so $x \in [b]$. Thus $[a] \subset [b]$. The reverse inclusion holds similarly.

- $(ii) \implies (iii)$: If $[a] = [b]$, then since R is reflexive, $a \in [a]$, so $a \in [b]$. Thus $a \in [a] \cap [b]$, so the intersection is not empty.

- $(iii) \implies (i)$: Assume $[a] \cap [b] \neq \emptyset$. Let $s \in [a] \cap [b]$. Then aRs and bRs. By symmetry sRb. By transitivity $aRs \wedge sRb \implies aRb$.

4.3 Partitions

A partition of a set S is a collection of disjoint non-empty subsets $S_1, \ldots, S_k$ such that $\bigcup S_i = S$. The set of equivalence classes forms a partition of S.

$$S = \bigcup_{x \in S} [x]$$

and distinct equivalence classes are disjoint.

4.4 Example: Modular Arithmetic

For $S = \mathbb{Z}$ and relation modulo 4:

- $[0] = \{\ldots, -8, -4, 0, 4, 8, \ldots\}$
- $[1] = \{\ldots, -7, -3, 1, 5, 9, \ldots\}$
- $[2] = \{\ldots, -6, -2, 2, 6, 10, \ldots\}$
- $[3] = \{\ldots, -5, -1, 3, 7, 11, \ldots\}$

These sets form a partition of $\mathbb{Z}$.

5 Partial Orders

5.1 Definition

A relation R on a set S is a **partial order** if it is:

1. Reflexive

2. Antisymmetric
3. Transitive

Notation: (S, R) or $(S, \leq)$ is called a **partially ordered set** (or **poset**).

5.2 Examples

1. $S = \mathbb{R}$, $xRy \iff x \leq y$. (Standard less-than-or-equal).
2. $S = \mathcal{P}(A)$ (Power set of A), $XRY \iff X \subseteq Y$. (Subset inclusion).
3. $S = \mathbb{N}$, $xRy \iff x|y$ (Divisibility).
   - Reflexive: $x|x$.
   - Antisymmetric: $x|y \wedge y|x \implies x = y$ (for positive integers).
   - Transitive: $x|y \wedge y|z \implies x|z$.
4. Lexicographic order (dictionary order) on strings. "bard" R "bark" because 'd' comes before 'k'.

5.3 Comparability

Let R be a partial order on S. Elements $a, b \in S$ are **comparable** if aRb or bRa. If every pair of elements in S is comparable, the partial order is called a **total** (or **linear**) order.

- Example of partial but not total: Subset inclusion. $\{1, 2\}$ and $\{2, 3\}$ are not comparable.
- Example of total order: $(\mathbb{R}, \leq)$.